

Synthetic Cultural Agents from Aggregate Anchors

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Can a sign-labeled adapter score a new survey item?

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Background

SCA 1.0 used cultural descriptions and persona-style prompting, and compared synthetic play to published tables from small-scale society experiments.

This paper changes the supplied information. Public GPS country z-scores become sign labels on a shared synthetic pair bank. One joint adapter per country is fitted with DPO. Evaluation prompts omit country names.

The lineage is simulation → an inspectable construction. We do not yet claim to have identified a preference parameter or a culture.

SCA 1.0: Capra, Gonzalez-Bonorino, and Pantoja. GPS: Falk et al. 2018. This draft: September 2026.

How much can simulation do on its own?

Human observations remain necessary. The open question is how many, and for what.

Prompted synthetic respondents mix the researcher's instructions with associations the model already has. That is useful for exploration. It is a weak basis for claiming that an item tracks a preference across populations.

A later human pilot is still the measurement step. The instrument we are discussing is a cheap preliminary readout whose value has to be tested.

Introduction and Discussion. Pre-pilot usefulness is prospective.

Problems documented, and the scope of this draft

Persona prompts and other inference-time strategies have many free knobs and mix several effects. Construct validity is then hard to audit. Most social-science uses of these models are about groups, while LLM-generated microdata are poorly understood. Human microdata remain expensive.

Scope. Synthetic comparisons; population-moment anchors; one DPO adapter per group, QLoRA, as a regularized Bradley–Terry pairwise choice fit.

Application. Transport between GPS anchors and WVS item text, compared with a country-named prompt. The question to keep in view: can the adapters score new survey items?

Revised paper §§1–2. No broad priority claim versus CultureLLM, Cao et al., SubPOP, or Jung and Kim.

Main research questions

- 1 Construction. Can we write an inspectable map from declared aggregate signs, through a shared pair bank, to a fitted scoring policy?
- 2 Transport. On held-out WVS text, with no country name in the prompt, do those policies retain GPS country order?
- 3 Agreement. Do they match WVS country orderings, and does that depend on the proposed item map?

Claim ladder: construction we have; compiler identification we do not; criterion association on trust order is partial.

From anchors to a synthetic dataset

Let $z \in \mathbb{R}^K$ collect K population moments. A labeling map sends z to a dataset of scenarios and opposing responses,

$$\mathcal{D} = \{(x, y^+, y^-)\}.$$

x is a scenario. y^+ is written to load high on a target facet; y^- to load low. Those loadings are generator judgments.

This first application cares about country *order*. The production map is the sign rule $s_k = 1\{z_k \geq 0\}$. Chosen = A when the target coordinate is non-negative, and B otherwise. Magnitudes are stored and later used as the GPS ranking in Spearman. They do not weight the DPO pairs.

Revised paper §3.1. $K = 6$: trust, risk, patience, altruism, positive reciprocity, negative reciprocity.

A sign profile is the construction unit

Argentina: $(-, +, -, +, +, -)$ trust / risk / patience / altruism / pos. recip. / neg. recip.

The United States is non-negative on all six coordinates, so every pair is labeled chosen A. Mexico is negative on all six, so every pair is labeled chosen B. That contrast is a global polarity flip of the same 658 scenarios.

16 countries yield 13 distinct sign profiles. Twins share a labeling problem: Mexico–Russia, India–Greece, Indonesia–Egypt.

Pause How should we use the twins as a robustness check?

If two adapters with the same s diverge on WVS, π is not a function of s alone. That test is unrun.

We replaced an LLM selector with the sign rule

The teacher and generator still write the shared A/B bank. An early labeling path asked a second model to pick A or B given a prose profile.

That selector was instructed that sign sets the direction. On the USA–Mexico checkpoint it matched the sign rule on 1,314 of 1,316 labels. The two misses are Mexico patience at $z = -0.11$.

Production labeling is therefore deterministic. Profile prose is unused. The 16 adapters train on $\text{sign}(z)$ only. The switch was to stop paying a model to re-implement a declared rule with noise.

Relabel run `gps_sign_relabel_all_20260816`. Default CLI: `deterministic_sign`.

DPO fits the labeled contests

$$\Pr(y_w \succ y_\ell \mid x) = \sigma\left(\beta \left[\log \frac{\pi_\theta(y_w \mid x)}{\pi_{\text{ref}}(y_w \mid x)} - \log \frac{\pi_\theta(y_\ell \mid x)}{\pi_{\text{ref}}(y_\ell \mid x)} \right]\right)$$

The loss raises the chosen text over the rejected text, relative to Llama 3.1 8B. The prompt-specific constant in the implicit reward cancels. This is a contest probability between two texts. It is not $\pi(y_w \mid x)$, and it is not the WVS code softmax.

USA: 658/658 chosen *A*. Mexico: 658/658 chosen *B*. The loss pushes $\sigma(\beta\Delta)$ toward 1 as a global polarity. Argentina is mixed: 328 *A* and 330 *B*, by target dimension. One joint adapter has to implement that patchwork.

Revised paper §3.2. $\beta = 0.1$, LoRA rank 16, one epoch, 526/132 split. Notebook settings.

We fit the whole bank jointly

All six signs enter the same QLoRA. A trust-item movement is therefore not an isolated trust-bit effect. Countries that differ on GPS trust also differ on other signs. Generator pairs load off-target facets.

What we can report is dimension-specific *transport*: on a named WVS map, do policy country ranks track that GPS z . What we cannot report is a ceteris paribus effect of one coordinate.

Llama 3.1 8B is retained because prompted survey simulation is known to be weak there. The test is whether a weight-embedded sign rule survives where a country name in the prompt is a poor instrument.

Revised paper §3.1. Per-dimension adapters are a later object.

Evaluation surface

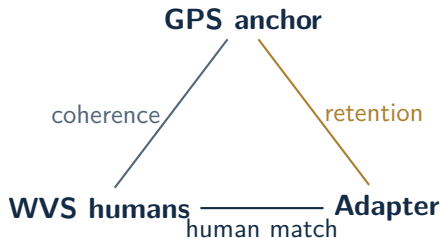
Sixteen purposive countries from the 42-country GPS–WVS intersection, chosen under compute constraints. Twenty-three single-choice WVS items on common support. Same recodes and aggregation across arms.

Egypt was not asked Q69–Q71; Britain was not asked Q174; those items are out everywhere. Patience Q43 and Q50 are scored separately.

Readout: teacher-forced log probability of each option code, leading space included, softmax over supplied codes only. No sampled answer. No country name in the primary prompt.

Table 2. Composites are candidate maps. Latent-scale validity remains to be assessed.

Three questions, one matched panel



Human–GPS: does this WVS map track GPS.

Adapter–GPS: does the unconditioned policy retain GPS order on new text.

Adapter–human: do policy ranks match people.

A country-named persona is a comparator.
It does not receive the sign vector.

Held-out synthetic pairs ask a prior question: did fitting recover the imposed labels.

Revised paper §3.3. Two correlations do not establish construct sharing.

Fitting recovers the imposed labels

0.942

Relative-reward recovery

Chosen–rejected margin improves over the reference.

0.819

Raw choice accuracy

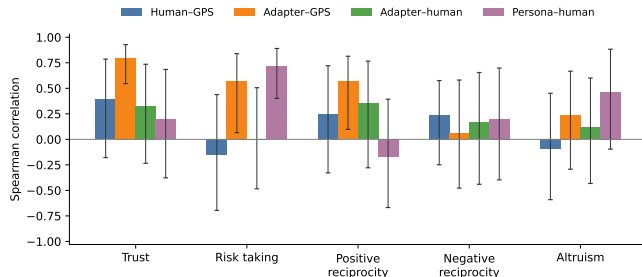
The adapter itself ranks chosen above rejected.

2,112 held-out country–pair cells. Russia uses a different 132-pair holdout.

Pause What independent test would make this more than a training metric?

Revised paper §4.1. Recovery is against synthetic labels.

Anchor retention and human agreement differ



Four series, different questions. Risk adapter-human has zero bar height.

Pause Which association should carry the paper's main empirical claim?

Trust retains GPS order; human agreement is uncertain

Country-rank association	ρ	95% interval
Adapter-GPS	0.80	[0.55, 0.93]
Persona-GPS	-0.07	[-0.55, 0.45]
Adapter-human	0.33	[-0.23, 0.74]

Paired adapter-minus-persona contrast: 0.87 [0.22, 1.36].

Permuting GPS trust z across frozen adapter scores, without refitting, gives one-sided $p = 0.002$.

Pause Is order-retention on this trust map enough for the contribution?

Revised paper §4.2. Intervals omit respondent, anchor, and fitting uncertainty.

A zero on risk does not settle the construct

Risk / adapter–human

$$\rho = 0$$

95% interval $[-0.48, 0.51]$

With 16 countries that interval includes large negative and large positive values.

Zero sample rank covariance is not independence of the response distributions, and it is not discriminant validity.

Patience / human–GPS

Q43: work importance -0.56

Q50: financial satisfaction $+0.62$

We keep the original coding and score the items separately.

Pause What evidence would separate a bad map from a policy that does not track people?

Table 4. Q43 interval $[-0.86, -0.04]$; Q50 $[0.11, 0.91]$.

Order and shape need not favor the same configuration

Configuration	Trust-GPS ρ	Mean TVD $\downarrow$
Base / unconditioned	—	0.462
Base / country persona	-0.07	0.369
Adapter / unconditioned	0.80	0.480
Adapter / country persona	0.67	0.401

The unconditioned adapter keeps the most GPS trust order and is farthest from human response distributions. A country-named base is the reverse.

On risk, adding a country name changes adapter-GPS from 0.57 to -0.26. These cells describe configuration sensitivity. They do not identify a general weights-versus-prompts law.

Table 5. TVD averages 368 country-item cells. Isolating route would require giving the prompt the same signs.

Main contributions of this draft

- 1 An inspectable construction: public GPS signs label a shared pair bank; one joint QLoRA-DPO adapter; country-unconditioned scoring.
- 2 An evaluation that keeps recovery, GPS-order transport, and human agreement as separate targets, with a persona comparator on the same items.
- 3 Evidence that, on this trust map, unconditioned adapters retain GPS country order more strongly than naming the country, while human ranks and distributional closeness remain unresolved and metric-dependent.

The object is a scoring instrument for group-level item screening. Independent human validation is still required before treating it as a pre-pilot.

Introduction contributions. Not identification of z , of a single coordinate, or of culture.

Open questions before a journal version

- 1 Second scenario bank, same GPS signs, refit two adapters.
- 2 Sign-retrain: permute the sixteen sign vectors, relabel, and refit.
- 3 Twin-profile exchangeability on held-out WVS.

The assignment permutation in the draft does not refit. Expanding 16 to 24 countries on this bank is later. A blinded human item-screening study is a use-test. It is not compiler identification.

Pause If we can only run one of these before submission, which one?

Paper 2. Egypt/Q69–Q71 already run: adapter–GPS trust stays 0.79–0.81.

What would you need to see before using this?

We can inspect the labeling rule and observe trust-order transfer on this panel.

Human agreement and practical pre-pilot value remain unresolved.

Discussion

Which part of the construction is least credible?

What evidence would change your view of the proposed use?

Which extension should we run first?

Claim ceiling: group-level scoring instruments intended to complement human survey research.

Appendix. A complete training pair

Scenario. A fellow student tutored me for free before a critical exam, and now they are facing a tight deadline for their own project and need someone to review their work. Decision: I can either take the time to thoroughly review their project or concentrate on my own study schedule. Trade-off: Helping my fellow student would acknowledge the effort they put into tutoring me, but it would cut into my own study time and potentially impact my performance.

A. I will set aside time to carefully review their work this evening. They went out of their way to help me when it mattered most, and I want to make sure they know their effort made a difference. If I do not take this seriously, it would feel like I am ignoring the support that helped me succeed.

B. I will let them know I can only do a quick scan tomorrow morning. My exam prep schedule is already packed with material I have not mastered yet, and I need to be realistic about what I can manage without burning out. I hope they understand my limitations in this situation.

Pair 00023ac74dac. Positive reciprocity. Argentina chooses A. Contractions expanded for typesetting.

Appendix. Candidate WVS indicators

Anchor	Items	Candidate interpretation
Trust	Q57–Q64, Q73	Generalized, interpersonal, institutional trust
Patience	Q43; Q50	Less work importance; financial satisfaction
Risk taking	Q106, Q107, Q109, Q178	Economic values; fare avoidance
Positive reciprocity	Q81	Confidence in charitable organizations
Negative reciprocity	Q176, Q177, Q179, Q195	Moral clarity; justifiability
Altruism	Q99, Q101, Q103	Membership in three types of organizations

Composites average item means without range standardization. Q57 human–GPS in this panel is 0.15.

Table 2. Strict panel excludes Q69–Q71, Q174, and Q12–Q14.

Appendix. How we score response codes

$$\ell_{c q j} = \sum_t \log \hat{\pi}_c(a_{q j t} \mid q, a_{q j, < t}), \quad \hat{P}_c(j \mid q) = \frac{e^{\ell_{c q j}}}{\sum_h e^{\ell_{c q h}}}$$

Candidate strings include a leading space. Sum token log probabilities without length normalization; normalize across the supplied codes only.

$$\text{Item score} = \sum_j g_q(j) \hat{P}_c(j \mid q)$$

Humans use the same recode before a survey-weighted mean. This softmax is a scoring convention. It is not the DPO contest probability.

Revised paper §3.3. Decoder-only model; no sampled response.

Appendix. Uncertainty, Egypt, mixed-trait bank

Country bootstrap, 2,000 draws, conditional on the 16 countries, items, and fitted runs. Intervals omit respondent sampling, GPS measurement error, and training randomness.

The shared bank is mixed-trait. Pair loadings are generator judgments. Egypt stays in the primary panel; restoring Q69–Q71 on the other fifteen countries leaves adapter–GPS trust between 0.79 and 0.81.

Still pending for public replay: pinned checkpoint provenance and an isolated calculation from published option probabilities.

Assignment permutation $p = 0.002$ on frozen scores. Sign-retrain is unrun.